

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Enterprise Java

M2:Practical –VIII

Roll No.: T002	Name: Asmitha Mohan Raj
Class: TYIT	Batch: 1
Date of Assignment: 29/08/2026	Date/Time of Submission: 29/08/2026

8. Implement the following EJB applications with different types of Beans:

a. Develop simple EJB application to demonstrate Servlet Hit count using Singleton Session Beans.

index.jsp:

```
<DOCTYPE html>
<html>
<head>
<title>Hit Count Demo</title>
</head>
<body>
<form action="ShowHitCountServlet"
method="post">
<input type="submit"
value="Take Hit Count demo with EJB Session
Singleton Bean"/>
```

ShowHitCountServlet.java:

```
//Add Imports :
package servlet;
import EJBPracts.STHitCountBean;
import java.io.IOException;
import java.io.PrintWriter;
import javax.ejb.EJB;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
//Add EJB Object :
@WebServlet("/ShowHitCountServlet")
public class ShowHitCountServlet extends HttpServlet {
    @EJB
    STHitCountBean cntBean;
//Add processRequest() :
protected void processRequest(HttpServletRequest request,
    HttpServletResponse response)
    throws ServletException, IOException {
    response.setContentType("text/html;charset=UTF-8");
}
```

Output:

Take Hit Count Demo with EJB Session Singleton Bean

You have visited this page: 1 time/s

```
</form>
```

```
</body>
```

```
</html>
```

STHitCountBean.java:

```
package EJBPracts;
import javax.ejb.Singleton;
@Singleton
public class STHitCountBean {
    private int count;
    public synchronized int incrHitCount() {
        return count++;
    }
    try (PrintWriter out = response.getWriter()) {
        out.println("<!DOCTYPE html>");
        out.println("<html>");
        out.println("<head>");
        out.println("<title>Hit Count</title>");
        out.println("</head>");
        out.println("<body>");
        out.println("<h1>You have visited this page: "
            + cntBean.incrHitCount()
            + " time/s</h1>");
        out.println("</body>");
        out.println("</html>");
    }
}
@Override
protected void doPost(HttpServletRequest request,
    HttpServletResponse response)
    throws ServletException, IOException {
    processRequest(request, response);
}
```

b. Develop simple visitor Statistics application using Message Driven Bean [MDB].**index.jsp:**

```

<%@page import="javax.jms.JMSEException"%>
<%@page import="javax.naming.InitialContext"%>
<%@page import="javax.jms.TextMessage"%>
<%@page import="javax.jms.MessageProducer"%>
<%@page import="javax.jms.Session"%>
<%@page import="javax.jms.Connection"%>
<%@page import="javax.jms.Queue"%>
<%@page import="javax.jms.ConnectionFactory"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
    <meta http-equiv="Content-Type"
        content="text/html; charset=UTF-8">
    <title>Message Driven Bean</title>
</head>
<body style="background-color:gray;">
<h1>Welcome To Message Driven Bean</h1>
<%
    ConnectionFactory connFact = null;
    Queue queue = null;
    Connection conn = null;
    Session mySess = null;
    MessageProducer msgProducer = null;
    TextMessage msg = null;
    try {
        InitialContext ctx = new InitialContext();
        queue = (Queue) ctx.lookup("jms/Queue");
        connFact = (ConnectionFactory)
            ctx.lookup("jms/QueueFactory");
        conn = connFact.createConnection();
        mySess = conn.createSession(
            false,
            Session.AUTO_ACKNOWLEDGE
        );
        msgProducer = mySess.createProducer(queue);
        msg = mySess.createTextMessage();
        msg.setText(request.getRemoteAddr());
        msgProducer.send(msg);
        conn.close();
    } catch (JMSEException e) {
        System.out.println(
            "Exception occurred: " + e.toString()
        );
    } catch (Exception e) {
        System.out.println(
            "Exception occurred: " + e.toString()
        );
    }
}%>
</body>
</html>

```

Sql:-

```

USE reg;
Database changed
CREATE TABLE visitorstat (
    visitdate TIMESTAMP NOT NULL,
    host VARCHAR(45),
    visitcount INT,
    PRIMARY KEY (host)
);

```

```
SELECT * FROM visitorstat;
```

MsgrvnBean.java:

```

package EJPract;
import javax.annotation.Resource;
import javax.ejb.ActivationConfigProperty;
import javax.ejb.EJB;
import javax.ejb.MessageDriven;
import javax.ejb.MessageDrivenContext;
import javax.jms.JMSEException;
import javax.jms.Message;
import javax.jms.MessageListener;
import javax.jms.TextMessage;
@MessageDriven(
    activationConfig = {
        @ActivationConfigProperty(
            propertyName = "destinationType",
            propertyValue = "javax.jms.Queue"
        ),
        @ActivationConfigProperty(
            propertyName = "destinationLookup",
            propertyValue = "jms/Queue"
        )
    }
)
public class MsgDrvnBean implements MessageListener {
    @EJB
    VStatelessBean vs;
    @Resource
    private MessageDrivenContext mdc;

    public MsgDrvnBean() {
    }
    @Override
    public void onMessage(Message message) {
        try {
            if (message instanceof TextMessage) {
                TextMessage msg =
                    (TextMessage) message;
                vs.addVisitor(msg.getText());
            }
        } catch (JMSEException e) {
            mdc.setRollbackOnly();
        }
    }
}

```

VStatelessBean.java:

```

package EJPract;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;
import javax.annotation.PostConstruct;
import javax.annotation.PreDestroy;
import javax.ejb.Stateless;
@Stateless
public class VStatelessBean {
    private Connection conn = null;
    private ResultSet rs;
}

```

```

private Statement stmt = null;
private String query = null;
@PostConstruct
public void connect() {
    try {
        Class.forName("com.mysql.jdbc.Driver");
        String connStr =
            "jdbc:mysql://localhost/reg";
        conn = DriverManager.getConnection(
            connStr,
            "root",
            "root"
        );

        System.out.println("Connection Success.");
    } catch (ClassNotFoundException | SQLException e) {
        System.err.println("Connection failed: " +
            e.getMessage());
    }
}

@PreDestroy
public void disconnect() {
    try {
        conn.close();
        System.out.println(
            "Connection closed."
        );
    }
    catch (SQLException e) {

        System.err.println(
            "Not able to close connection: "
            + e.getMessage()
        );
    }
}

public void addVisitor(String host) {
    try {

```

```

        stmt = conn.createStatement();
        query =
            "INSERT INTO visitorstat(host,visitcount) " +
            "VALUES('" + host + "',1)";

        stmt.executeUpdate(query);
    }
    catch (SQLException e) {
        try {
            stmt = conn.createStatement();
            query =
                "UPDATE visitorstat " +
                "SET visitcount = visitcount + 1 " +
                "WHERE host = '" + host + "'";
            stmt.executeUpdate(query);
        }
        catch (SQLException ex) {

            System.err.println(
                "Cannot update: "
                + ex.getMessage()
            );
        }
    }
}
}
}

```

Output:**Welcome To Message Driven Bean**

```

mysql> USE reg;
Database changed
mysql> SELECT * FROM visitorstat;
+-----+-----+-----+
| visitdate | host | visitcount |
+-----+-----+-----+
| 2026-08-29 11:24:28 | 0:0:0:0:0:0:1 | 18 |
+-----+-----+-----+
1 row in set (0.00 sec)

```

c. Develop simple Marks Entry Application to demonstrate accessing Database using EJB.**index.jsp:**

```

<%@page import="java.util.ArrayList"%>
<%@page import="java.util.Iterator"%>
<%@page import="java.util.List"%>
<%@page import="javax.naming.InitialContext"%>
<%@page import="EJPract.MarksEntryBean"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<%!
    private static MarksEntryBean marksEntry;
    public void jspInit() {
        try {
            InitialContext ic = new InitialContext();
            marksEntry = (MarksEntryBean)
                ic.lookup(
                    "java:global/EJBMarksEntry/MarksEntryBean"
                );
        }
        catch (Exception ex) {
            System.out.println(
                "Database Error " + ex.getMessage()
            );
        }
    }
}

```

```

%>
<%
    if (request.getParameter("insertMarks") != null) {
        String name, marks1, marks2, marks3;
        name = request.getParameter("stuName");
        marks1 = request.getParameter("sub1");
        marks2 = request.getParameter("sub2");
        marks3 = request.getParameter("sub3");
        marksEntry.addMarks(
            name,
            marks1,
            marks2,
            marks3
        );
    }
%>
<html>
<head>
    <meta http-equiv="Content-Type"
        content="text/html; charset=UTF-8">
    <title>Marks Entry</title>
</head>
<body>
    <h2 style="text-align:center;">
        Enter Details

```

```

</h2>
<br>
<form name="result" method="post">
<table>
<tr>
<td>Enter Students Name</td>
<td>
<input type="text"
name="stuName"
value="">
</td>
</tr>
<tr>
<td>Enter Subject 1 Marks</td>
<td>
<input type="text"
name="sub1"
value="">
</td>
</tr>
<tr>
<td>Enter Subject 2 Marks</td>
<td>
<input type="text"
name="sub2"
value="">
</td>
</tr>
<tr>
<td>Enter Subject 3 Marks</td>
<td>
<input type="text"
name="sub3"
value="">
</td>
</tr>
<tr>
<td colspan="2">
<input type="submit"
value="insertMarks"
name="insertMarks">
</td>
</tr>
</table>
</form>
</body>
</html>

```

MarksEntryBean.java:

```

package EJPract;
import java.sql.*;
import java.util.ArrayList;
import java.util.List;
import javax.ejb.Stateful;
@Stateful
public class MarksEntryBean {
    List<String> details;
    String stuName, sub1, sub2, sub3;
    private Connection conn = null;
    private ResultSet rs;
    private Statement stmt = null;
    private String query = null;
    public void createConnection() {
        try {
            Class.forName("org.gjt.mm.mysql.Driver");

```

```

String connStr =
    "jdbc:mysql://localhost/reg";
conn = DriverManager.getConnection(
    connStr,
    "root",
    "root"
);
} catch (ClassNotFoundException | SQLException e) {
    System.err.println(
        "Database Error " + e.getMessage()
    );
}
}
}
public void addMarks(
    String stuname,
    String marks1,
    String marks2,
    String marks3) {
    createConnection();
    try {
        stmt = conn.createStatement();
        query =
            "INSERT INTO marks" +
            "(stuName,sub1,sub2,sub3)" +
            "VALUES(" +
            stuname + "," +
            marks1 + "," +
            marks2 + "," +
            marks3 + ")";
        stmt.executeUpdate(query);
    }
    catch (SQLException e) {
        System.err.println(
            "Failed to insert. "
            + e.getMessage()
        );
    }
}
}
}
}

```

Output:

```

mysql> SELECT * FROM marks;
+----+-----+-----+-----+-----+
| id | stuName | sub1 | sub2 | sub3 |
+----+-----+-----+-----+-----+
| 1  | aashu   | 100  | 69   | 100   |
| 2  | Asmitha | 100  | 100  | 100   |
| 3  | Afsha   | 100  | 100  | 100   |
| 4  | Radhika | 1000 | 100  | 100   |
+----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

```